

Transformers

TRANSFORMERS CHANGE THE VOLTAGE OF AC POWER.

Alternating current can be changed to a higher or lower voltage by a device called a transformer. For example, high voltage from a power station needs to be transformed to a lower voltage for use in homes.

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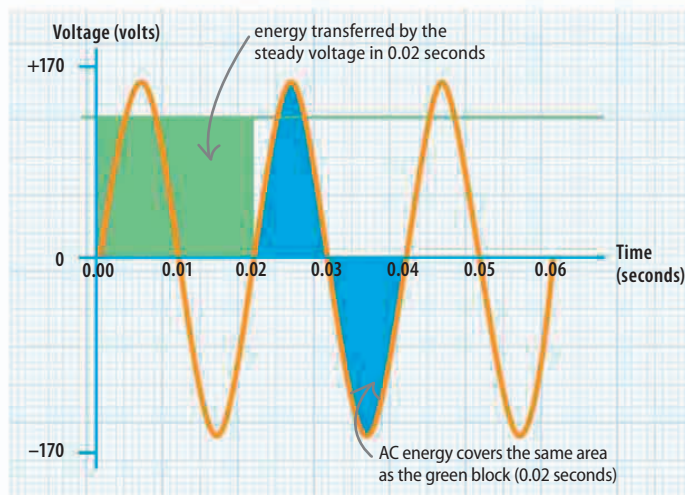
220 ▶

Direct and alternating current

There are two kinds of current: direct and alternating. Direct current (DC) is usually produced by batteries and it flows one way around a circuit. Electricity—as used in homes—has an alternating current (AC), in which the direction of the flow of electricity reverses dozens of times a second. Transformers are devices that can be used to change the voltages and currents of AC (see pages 204–205). They make it easy to change AC to a high-voltage form for transmission over long distances, and to a low-voltage form for domestic use.

▷ AC and DC voltage

In this graph, the green line represents DC voltage and the green area is the energy transferred by this voltage. To transfer the same energy in the same time (as shown by the blue areas), the AC voltage must rise higher than the DC voltage at some parts of its cycle, as the orange line shows.



Transformers

An inductor is a coil of wire that stores energy in a magnetic field. A transformer is two inductors in one: two coils share the same core. When an alternating current passes through one coil, the core sets up currents in the other coil. If this second coil has more turns than the first, then the voltage across it is higher.

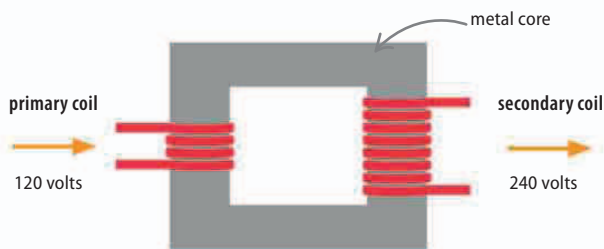
▷ Voltage and coils

The ratio of the number of volts that pass through the primary and secondary transformers is equal to the ratio of the number of turns in the primary and secondary coils.

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

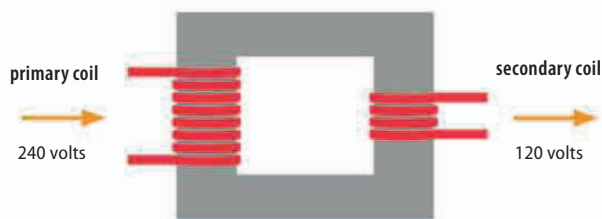
Labels for the equation:

- V_p : voltage (volts) across primary coil
- V_s : voltage (volts) across secondary coil
- N_p : number of turns of primary coil
- N_s : number of turns of secondary coil



△ Step-up transformer

The second inductor has twice as many turns as the first, so its voltage is twice that in the first.



△ Step-down transformer

The second inductor has half as many turns as the first, so the voltage is halved as a result.